

Apache Cassandra

3-Node Cluster Setup Guide

Platform: VMware Workstation / vSphere

OS: Rocky Linux 8/9

Cassandra Version: 4.1.x

1. Overview & Architecture

This guide walks you through a complete step-by-step configuration of a 3-node Apache Cassandra cluster running on Rocky Linux virtual machines hosted on VMware. Each section covers one phase of the setup, from VM provisioning to cluster verification.

Cluster Architecture

Node	Hostname	IP Address	Role	RAM	CPU	Disk
Node 1	cassandra-node1	192.168.1.101	Seed	8 GB	4 vCPU	100 GB
Node 2	cassandra-node2	192.168.1.102	Seed	8 GB	4 vCPU	100 GB
Node 3	cassandra-node3	192.168.1.103	Non-Seed	8 GB	4 vCPU	100 GB

i NOTE

Adjust IP addresses to match your VMware network segment. The IPs above assume a /24 subnet. Change them consistently in every configuration file.

Key Concepts

- **Seed Nodes:** Bootstrap nodes that help new nodes discover the cluster. At least 2 seeds recommended.
- **Snitch:** Determines the topology. `GossipingPropertyFileSnitch` is used for production.
- **Replication Factor:** Set to 3 (one copy per node) so data survives up to 2 node failures.
- **Consistency Level:** `QUORUM` (2 of 3 nodes must acknowledge) balances consistency and availability.

2. VMware VM Provisioning

Create three identical VMs in VMware Workstation or vSphere. Repeat the steps below for each node.

2.1 VM Specifications

Guest OS	Rocky Linux 8 or Rocky Linux 9 (64-bit)
vCPUs	4 (minimum 2)
RAM	8 GB (minimum 4 GB for testing)

Disk	100 GB thin-provisioned — separate data disk recommended
Network Adapter	VMXNET3 — set to Bridged or Host-only depending on your lab
VM Tools	Install open-vm-tools after OS installation

2.2 Network Configuration

Set static IPs on each VM. Using DHCP risks IP changes on reboot which breaks the cluster.

```
# On Node 1 — edit the network connection
sudo nmcli con mod ens192 ipv4.addresses 192.168.1.101/24
sudo nmcli con mod ens192 ipv4.gateway 192.168.1.1
sudo nmcli con mod ens192 ipv4.dns 8.8.8.8
sudo nmcli con mod ens192 ipv4.method manual
sudo nmcli con up ens192

# Repeat on Node 2 with 192.168.1.102 and Node 3 with 192.168.1.103
```

✓ TIP

Replace ens192 with your actual NIC name (check with: `ip link show`). VMware VMs typically use ens160 or ens192.

2.3 Hostname Setup

```
# Node 1
sudo hostnamectl set-hostname cassandra-node1

# Node 2
sudo hostnamectl set-hostname cassandra-node2

# Node 3
sudo hostnamectl set-hostname cassandra-node3
```

Add all nodes to `/etc/hosts` on every VM so they can resolve each other:

```
sudo tee -a /etc/hosts <<EOF
192.168.1.101 cassandra-node1
192.168.1.102 cassandra-node2
192.168.1.103 cassandra-node3
EOF
```

3. OS Preparation (All 3 Nodes)

Perform the following steps on all three nodes unless stated otherwise.

3.1 System Update

```
sudo dnf update -y
sudo dnf install -y epel-release
sudo dnf install -y wget curl tar net-tools open-vm-tools
sudo systemctl enable vmtoolsd --now
```

3.2 Disable Swap

Cassandra performs poorly with swap enabled. Disable it permanently.

```
# Disable immediately
sudo swapoff -a

# Disable permanently (comment out swap line)
sudo sed -i '/ swap / s/^#/' /etc/fstab

# Verify swap is off
free -h # Swap line should show 0
```

3.3 Configure Firewall

Open the ports required by Cassandra on all nodes:

7000	Inter-node cluster communication (un-encrypted)
7001	Inter-node TLS communication (if TLS enabled)
7199	JMX monitoring port
9042	Native CQL client port
9160	Thrift client port (legacy, optional)

```
# Allow Cassandra ports
sudo firewall-cmd --permanent --add-port=7000/tcp
sudo firewall-cmd --permanent --add-port=7001/tcp
sudo firewall-cmd --permanent --add-port=7199/tcp
sudo firewall-cmd --permanent --add-port=9042/tcp
sudo firewall-cmd --permanent --add-port=9160/tcp

# Reload firewall
sudo firewall-cmd --reload
```

```
# Verify
sudo firewall-cmd --list-ports
```

3.4 Disable SELinux (or configure it)

⚠ WARNING

Disabling SELinux is the quickest approach for a lab. In production, configure proper SELinux policies instead.

```
# Disable SELinux temporarily
sudo setenforce 0

# Disable permanently — edit config file
sudo sed -i 's/^SELINUX=enforcing/SELINUX=permissive/' /etc/selinux/config

# Verify
getenforce    # Should output: Permissive
```

3.5 Configure OS Limits for Cassandra

Cassandra requires higher open file and process limits. Add the following:

```
sudo tee /etc/security/limits.d/cassandra.conf <<EOF
cassandra - nofile    100000
cassandra - nproc     32768
cassandra - memlock   unlimited
EOF
```

```
# Also set kernel parameters
sudo tee /etc/sysctl.d/99-cassandra.conf <<EOF
vm.max_map_count=1048575
net.core.rmem_max=16777216
net.core.wmem_max=16777216
EOF

sudo sysctl --system
```

4. Java Installation (All 3 Nodes)

Apache Cassandra 4.x requires OpenJDK 11. Cassandra 5.x requires OpenJDK 11 or 17.

4.1 Install OpenJDK 11

```
sudo dnf install -y java-11-openjdk java-11-openjdk-devel
```

```
# Verify installation
java -version
# Expected output: openjdk version "11.0.x"
```

4.2 Set JAVA_HOME

```
# Find the Java path
which java
readlink -f $(which java)

# Add to /etc/environment
echo 'JAVA_HOME=/usr/lib/jvm/java-11-openjdk' | sudo tee -a /etc/environment
source /etc/environment

# Verify
echo $JAVA_HOME
```

5. Cassandra Installation (All 3 Nodes)

5.1 Add Apache Cassandra Repository

```
# Create the repo file
sudo tee /etc/yum.repos.d/cassandra.repo <<EOF
[cassandra]
name=Apache Cassandra
baseurl=https://redhat.cassandra.apache.org/41x/
gpgcheck=1
repo_gpgcheck=1
gpgkey=https://downloads.apache.org/cassandra/KEYS
EOF
```

✓ TIP

For Cassandra 4.0.x change 41x to 40x in the baseurl. Check <https://cassandra.apache.org/download/> for the latest repo paths.

5.2 Install Cassandra

```
sudo dnf install -y cassandra

# Verify installed version
cassandra -v
```

5.3 Directory Structure

The default Cassandra directories on Rocky Linux after installation:

Config files	/etc/cassandra/conf/
Data directory	/var/lib/cassandra/data/
Commit log	/var/lib/cassandra/commitlog/
Hints	/var/lib/cassandra/hints/
Log files	/var/log/cassandra/
Binary tools	/usr/bin/cqlsh, nodetool, etc.

✓ TIP

For production, mount data and commitlog on separate fast disks (NVMe SSDs). Use VMware Raw Device Mapping (RDM) or thick-provisioned VMDK for best I/O performance.

6. Cassandra Configuration

This is the most critical section. The `cassandra.yaml` file must be configured differently on each node — only a few parameters differ between nodes.

6.1 Common Parameters (All 3 Nodes)

Edit `/etc/cassandra/conf/cassandra.yaml` on each node:

```
sudo vi /etc/cassandra/conf/cassandra.yaml
```

Set the following parameters (they are the same on all nodes):

```
# Cluster name - MUST be identical on all nodes
cluster_name: 'MyProductionCluster'

# Snitch - use GossipingPropertyFileSnitch for multi-DC
endpoint_snitch: GossipingPropertyFileSnitch

# Seed nodes - list BOTH seed IPs
seed_provider:
  - class_name: org.apache.cassandra.locator.SimpleSeedProvider
    parameters:
      - seeds: "192.168.1.101,192.168.1.102"

# Authenticator and Authorizer (enable for production)
authenticator: PasswordAuthenticator
authorizer: CassandraAuthorizer

# Partitioner (default - do not change for new clusters)
partitioner: org.apache.cassandra.dht.Murmur3Partitioner

# Data file directories
data_file_directories:
  - /var/lib/cassandra/data

commitlog_directory: /var/lib/cassandra/commitlog
hints_directory: /var/lib/cassandra/hints
saved_caches_directory: /var/lib/cassandra/saved_caches

# Enable JMX for nodetool
LOCAL_JMX=yes
```

6.2 Node-Specific Parameters

The following parameters are DIFFERENT on each node:

Parameter	Node 1	Node 2	Node 3
listen_address	192.168.1.101	192.168.1.102	192.168.1.103
rpc_address	192.168.1.101	192.168.1.102	192.168.1.103
broadcast_address	192.168.1.101	192.168.1.102	192.168.1.103

Add/update the node-specific lines in `cassandra.yaml`:

```
# ---- Node 1 (cassandra-node1) specific settings ----
listen_address: 192.168.1.101
rpc_address: 192.168.1.101
broadcast_address: 192.168.1.101
broadcast_rpc_address: 192.168.1.101

# ---- Node 2 (cassandra-node2) specific settings ----
listen_address: 192.168.1.102
rpc_address: 192.168.1.102
broadcast_address: 192.168.1.102
broadcast_rpc_address: 192.168.1.102

# ---- Node 3 (cassandra-node3) specific settings ----
listen_address: 192.168.1.103
rpc_address: 192.168.1.103
broadcast_address: 192.168.1.103
broadcast_rpc_address: 192.168.1.103
```

6.3 Configure `cassandra-rackdc.properties`

Required when using `GossipingPropertyFileSnitch`. Edit on all 3 nodes:

```
sudo vi /etc/cassandra/conf/cassandra-rackdc.properties

# Set datacenter and rack (same for single-DC lab setup)
dc=DC1
rack=RACK1
```

6.4 JVM Heap Settings

Edit the JVM options file to set appropriate heap memory (typically 1/4 to 1/2 of RAM):

```
sudo vi /etc/cassandra/conf/jvm-server.options

# For 8 GB RAM - set heap to 2 GB
-Xms2G
-Xmx2G

# For 16 GB RAM - set heap to 4 GB
-Xms4G
-Xmx4G
```

⚠ WARNING

Never exceed 31 GB for heap — JVM compressed OOPs are disabled above that threshold which degrades performance significantly.

7. Starting the Cluster

⚠ WARNING

Always start SEED NODES first. Wait 30–60 seconds between starting each node.

7.1 Start Node 1 (Seed)

```
# Enable and start Cassandra on Node 1
sudo systemctl enable cassandra
sudo systemctl start cassandra

# Watch the startup log
sudo tail -f /var/log/cassandra/system.log

# Wait until you see this message before proceeding:
# INFO [main] ... Starting listening for CQL clients on /192.168.1.101:9042
```

7.2 Start Node 2 (Seed)

Wait at least 30 seconds after Node 1 is UP, then on Node 2:

```
sudo systemctl enable cassandra
sudo systemctl start cassandra
sudo tail -f /var/log/cassandra/system.log
```

7.3 Start Node 3

After Node 2 is stable (30–60 seconds), start Node 3:

```
sudo systemctl enable cassandra
sudo systemctl start cassandra
sudo tail -f /var/log/cassandra/system.log
```

8. Cluster Verification

8.1 Check Cluster Status

Run nodetool status from any node. All nodes should show UN (Up/Normal):

```
nodetool status

# Expected output:
Datacenter: DC1
=====
Status=Up/Down
```

```
|/ State=Normal/Leaving/Joining/Moving
--  Address          Load        Tokens   Owns    Host ID      Rack
UN  192.168.1.101    256.23 KiB   16       33.3%   <uuid>      RACK1
UN  192.168.1.102    283.91 KiB   16       33.3%   <uuid>      RACK1
UN  192.168.1.103    271.05 KiB   16       33.3%   <uuid>      RACK1
```

⚠ WARNING

If a node shows DN (Down/Normal) or UJ (Up/Joining), check the system.log on that node for errors. Common causes: wrong IP in cassandra.yaml, firewall blocking port 7000, or clock skew between VMs.

8.2 Additional nodetool Checks

```
# Check ring topology
nodetool ring

# View gossip information
nodetool gossipinfo

# Check if all nodes are participating
nodetool describecluster

# Check data distribution
nodetool tpstats
```

8.3 Connect with cqlsh and Test

```
# Connect to the local node
cqlsh 192.168.1.101 9042

# Or with authentication (if PasswordAuthenticator is enabled)
cqlsh 192.168.1.101 9042 -u cassandra -p cassandra

# Once connected — create a test keyspace with replication factor 3
CREATE KEYSPACE test_ks
WITH replication = {
    'class': 'NetworkTopologyStrategy',
    'DC1': 3
};

# Create a test table
USE test_ks;
CREATE TABLE users (
    user_id UUID PRIMARY KEY,
    name TEXT,
    email TEXT
);
```

```
# Insert data
INSERT INTO users (user_id, name, email)
  VALUES (uuid(), 'Test User', 'test@example.com');

# Query data
SELECT * FROM users;

# Verify data is replicated on all 3 nodes
nodetool getendpoints test_ks users <partition-key-value>
```

9. Post-Configuration & Security

9.1 Change Default Cassandra Password

⚠ WARNING

The default `cassandra/cassandra` superuser must be changed immediately on any non-lab system.

```
# Connect as default cassandra user
cqlsh 192.168.1.101 -u cassandra -p cassandra

# Create a new superuser
CREATE ROLE admin WITH SUPERUSER = true
    AND LOGIN = true
    AND PASSWORD = 'StrongPassword123!';

# Log in as admin and drop the default user
# (Reconnect first as admin)
DROP ROLE cassandra;
```

9.2 Enable Inter-node Encryption (Optional but Recommended)

Generate SSL certificates and configure `server_encryption_options` in `cassandra.yaml` on all nodes:

```
# In cassandra.yaml - enable inter-node encryption
server_encryption_options:
    internode_encryption: all
    keystore: /etc/cassandra/conf/.keystore
    keystore_password: cassandra
    truststore: /etc/cassandra/conf/.truststore
    truststore_password: cassandra
```

9.3 Configure VMware Snapshots

Take VMware snapshots before and after configuration milestones:

- Take a snapshot after OS setup and before Cassandra install
- Take a snapshot after Cassandra install and before starting the cluster
- Take a snapshot once the 3-node cluster is verified healthy
- Quiesce the guest OS (use VMware snapshot with memory if Cassandra is running)

9.4 NTP / Time Synchronization

⚠ WARNING

Cassandra is time-sensitive. All nodes **MUST** have synchronized clocks. Time drift causes write conflicts and tombstone issues.

```
# Install and enable chrony (NTP) on all nodes
sudo dnf install -y chrony
sudo systemctl enable chronyd --now

# Verify time sync
chronyc tracking
chronyc sources -v

# Check time difference between nodes (should be < 500ms)
date      # Run on each node and compare
```

10. Troubleshooting Reference

Issue / Symptom	Resolution
Node stuck in UJ (Joining) state	Another node may be bootstrapping simultaneously. Restart the joining node after seeds are fully UP. Check system.log for errors.
Connection refused on port 9042	Check firewall with: firewall-cmd --list-ports. Ensure Cassandra is running: systemctl status cassandra.
nodetool error: Connection refused to 7199	JMX is not running or is bound to wrong address. Ensure LOCAL_JMX=yes in cassandra-env.sh or set JVM_OPTS for JMX host.
Nodes cannot see each other	Verify seed IPs are correct. Check firewall allows port 7000. Test: telnet 192.168.1.102 7000 from node1.
Cassandra fails to start (OOM)	Heap is too large. Reduce Xmx in jvm-server.options. Check: journalctl -u cassandra -n 50.
Schema disagreement warning	Temporarily normal during node joins. If persistent, run: nodetool describecluster and check schema versions.
Clock skew detected in logs	Sync NTP on all nodes: chronyc makestep. Ensure all VMs point to the same NTP server.
Permission denied on data directories	Fix ownership: chown -R cassandra:cassandra /var/lib/cassandra /var/log/cassandra

Useful Log Commands

```
# Live Cassandra system log
sudo tail -f /var/log/cassandra/system.log

# Systemd journal
sudo journalctl -u cassandra -f

# Check Cassandra process
ps aux | grep cassandra

# Check open ports
ss -tlnp | grep -E '7000|7001|9042|7199'
```


11. Quick Reference Card

Key File Locations

Main config	/etc/cassandra/conf/cassandra.yaml
Rack/DC config	/etc/cassandra/conf/cassandra-rackdc.properties
JVM settings	/etc/cassandra/conf/jvm-server.options
Log4j config	/etc/cassandra/conf/logback.xml
System log	/var/log/cassandra/system.log
Data files	/var/lib/cassandra/data/
Commit log	/var/lib/cassandra/commitlog/

Essential Commands

```
# Service management
sudo systemctl start|stop|restart|status cassandra

# Cluster health
nodetool status
nodetool ring
nodetool describecluster

# Node operations
nodetool drain           # Flush memtables before stopping
nodetool decommission    # Remove node from cluster gracefully
nodetool repair          # Repair data inconsistencies

# Connect to CQL
cqlsh <node-ip> 9042
cqlsh <node-ip> 9042 -u <user> -p <pass>

# Performance
nodetool tpstats          # Thread pool stats
nodetool compactionstats  # Running compactions
nodetool cfstats          # Per-table statistics
```

End of Document

Apache Cassandra 3-Node Cluster — Rocky Linux on VMware